

2 ENERGY CORRELATORS AS A PROBE OF THE HARD PROCESS IN RUN 24 $p - p$
3 COLLISIONS AT $\sqrt{s} = 200 \text{ GeV}$ WITH THE sPHENIX DETECTOR AT RHIC

by

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Abstract

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Adviser: Professor Stefan Bathe

This dissertation consists of three parts in addition to a literature review establishing the state of the art of jet physics and the Energy-Energy correlator at high energies...

Part 1: Hardware This part discusses the physical hardware used to make the measurements of the energy in the jets created by the proton-proton collisions. This part contains a discussion of the sPHENIX detector and an in-depth look at the relevant subsystems for the measurements in this analysis. In addition, the Monte Carlo methods and models to discern the expected response and calibrate the detector are discussed in detail, with both subparts coming together in a discussion of backgrounds and error calculation in general and for the observables at hand.

Part 2: Technicalities This part builds from first principles up to the jet measurement in a theoretical light, and then discusses the additional computational techniques on display that will provide the bridge between the theory of these observables and the practical application to sPHENIX. This part contains a chapter that is a pared down discussion of a midstream paper proposed as part of the thesis process that establishes the safety of this observable against jet finding choices.

39 **Part 3: Experimental Output** This part is the meat and potatoes of the dissertation,
40 discussing results from the experiment and the analysis in light of the previous parts and
41 comparing to the world results from related systems.

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123 Chapter 1

124 Literature Review

1.1 Jets Definitions

The study of Quantum Chromodynamics in high energy collisions, such as those at the Relativistic Heavy Ion Collider or the Large Hadron Collider, is often carried out through investigation of the kinematics of jets. Jets, as objects, sit in the boundary between experiment and theory, being an experimental signature corresponding to final states of quarks and gluons produced in collisions. Jets are a cluster of final state particles that result from the showering and hadronization of the initial parton, that are identified in experiment through use of one of the multiple jet reconstruction algorithms.

1.1.1 Jet Reconstruction Algorithms

In Ryan Atkin's paper *Review of jet reconstruction algorithms* [1], Atkin provides an overview of the standard algorithms with a focus on practical implementation for experimental usage. Atkin discusses a number of algorithms that fall into two categories: Cone based and Sequential Clustering

Cone Methods

In Atkin's paper, Atkin discusses two iterative cone procedures—Iterative Cone with Progressive Removal (IC-PR) and with Split Merge (IC-SM)—and one more modern cone method, Seedless Infrared Safe Cone (SIS-Cone). These iterative cone procedures are described as simple to implement in an experimental context, however, they both suffer deficiencies from a theoretical perspective as they fail to be IRC safe [1] [2]. In principle, these clustering algorithms are rather similar, one takes the hardest particle in the final state, terms this a jet seed and draws a circle of a fixed size around it, summing the 4-momentum of all particles of within the circle. Then, the jet axis is compared to the jet seed, where a properly identified

147 cone is defined as one where the condition in eq (1.1) is fulfilled.

$$\frac{p_{\text{jet}}^\mu}{|p_{\text{jet}}|} = \frac{p_{\text{seed}}^\mu}{|p_{\text{seed}}|} \quad (1.1)$$

148 If this condition is not fulfilled, one then starts again, setting the seed at the axis of the
 149 previously identified cone and draws a new circle of the same radius centered there. Once
 150 the jet is identified, the progressive removal procedure then removes all particles identified
 151 with this method and continues until there are no particles with a $p_T > p_T^{\text{min}}$ which is set by
 152 the user. In the split-merge procedure, one first identifies all seed particles, and builds the
 153 stable cones, but these are then treated as protojets. These protojets are then filtered by an
 154 additional $p_T^{\text{jet}} > p_T^{\text{cut}}$ threshold, then overlapping protojets are split or merged in descending
 155 p_T order depending on a merging parameter, f 0.5. If the sum of p_T for the shared particles
 156 is less than the $f p_T^{\text{sl}}$ where p_T^{sl} is the lower p_T protojet, then the overlapping particles are
 157 split based on distance from the protojet axes and the protojets are updated. Else wise, the
 158 two protojets are merged into a single jet.

159 Both of these methods are IRC unsafe to some degree, but are useful in situations where
 160 quick jet identification is needed, e.g. CMS Cone and Pythia Cone are IC-PR and ATLAS
 161 Cone and JetClu midpoint cones are IC-SM. The IC-PR is collinear unsafe, as the progressive
 162 removal procedure inherently ignores any overlapping jets. The IC-SM however is infrared-
 163 unsafe as it introduces further parameters to filter pile-up and thus removes low energy
 164 contributions [2].

165 Sequential Clustering algorithms

166 The sequential clustering algorithms on the other hand are all IRC safe, working from the
 167 underlying assumption that all of the final state particles in a jet would have originated from
 168 a single initial parton, thus in theory, one can reconstruct that collection [3]. Three of these

methods, anti- k_T , k_T and Cambridge/Aachen, utilize the same mechanism with modifications to their metric. These methods, here referred to as k_T -type, rely on the evaluation of a distance metric of the form,

$$d_{ij} = \min \{p_{T,i}^{2n}, p_{T,j}^{2n}\} \frac{\Delta R_{ij}}{R} \quad d_i = p_{T,i}^{2n} \quad (1.2)$$

Where $n = 1$ is k_T , $n = 0$ is the Cambridge/Aachen and $n = -1$ is the anti- k_T , ΔR_{ij} is the angular distance between the jets in $\phi - y(\eta)$ space and R is the size of the jet, which functions as an estimated value of the average jet radius, a parameter set by the analyzer. One clusters particles together by finding $\min \{\{d_{ij}\}, \{d_i\}\}$ and, if the minimum value is one of these d_{ij} then one clusters those two, adds their 4 momentum together and recalculates the distance metric. If the minimum is a d_i —called beam distance, then the protojet is promoted to a jet and then removed from the set of candidates. While these methods are IRC safe [3], they do suffer from a similar tuning problem as the cone methods, where misidentification of jets is possible depending on the R value and p_T cutoff, however, both of these parameters can be estimated well and have led to a standard prescription of the anti- k_T method at a variety of R , usually prioritizing the $R = 0.4$ jet as the best balance between exclusion and false positive jet particles. Of note for the analysis of this system, the k_T -type algorithms can produce different collections of jet particles and different shapes of the final state of the jet. The anti- k_T method produces circular cones as a function of its metric, thus making it more directly comparable to the previous cone methods, while the k_T method will produce final states of various sizes. This discrepancy proves useful in analysis of jet substructure observables, such as the primary family of observables—the Energy-Energy Correlators—in this thesis, as it allows for a Lund Plane analysis of the jet structure that combines anti- k_T and k_T techniques to improve veracity of the analysis method

1.1.2 Quark and Gluon Jets

Standard Definitions: Event level label

Experimentally, there is significant interest in being able to discriminate quark and gluon jets in searches for Beyond Standard Model physics, searches for Dark Matter, in jet substructure analysis and in QGP physics. [4][5][6][7]. To first order, the question of a theoretical definition is seemingly trivial. In *How to see the differences between quark and gluon jets*[7], Fodor posits a model independent mechanism to distinguish between quark and gluon jets in $e^+e^- \rightarrow jj + X$ events. As one would expect, Fodor notes that the underlying assumption is that "jets are the footprints of the partons....Thus we should determine the parton ancestor of a jet" [7]. From here, Fodor goes on to note significant distinguishing characteristics of the parton populations, noting the gluon has a stronger color charge and is of spin-1 as compared to the spin-half quark, and that this color charge is reflected in the comparative fragmentations, inducing higher multiplicity, softer hadron spectrum and a broader p_T spectrum. To this end, Fodor focuses on the three jet topology as the e^+e^- dijet is exclusively populated by $q\bar{q}$ topology, however, in the three jet case, one introduces $e^+e^- \rightarrow q\bar{q}g$ type events. Fodor notes that previous attempts to study this ratio have been hampered by a lack of consideration of jet energy and insufficient statistics. Further, Fodor notes that the observable of interest must suppress the "'problematic' soft particle [contribution]" and be dependent on the energy of the jet.

It is here where we get our first inklings of the difficulties of the "parton-ancestor" jet definition, which while tempting suffers from a non-physicality at the point of hadronization wherein a color singlet or doublet state goes to a color-neutral state, hence problematizing the simple "evolution" model of a jet. One can not draw a Feynman diagram of a jet, nor will any combination of quark and gluon vertexes ever lead to a final state with no free partons present. This deficiency is even seen in the observational challenges noted by Fodor,

where, the assigning of soft particles to one jet or the other becomes a problematic task for the analyzer and the particular jet algorithm and energy must be carefully tuned for the two styles. It is these deficiencies that, while not insurmountable, would lead one to seek a more phenomenologically pleasing definition.

Alternate Definition: Sample Populations

In their paper, *An Operational Definition of Quark and Gluon Jets* [8] Kimiske, Metodiev and Thaler present a theoretical definition of jet flavor with an eye towards application at colliders. They aim to present a precise and practical definition of jet flavor at the hadron level, thus allowing for direct measurement from collider experiments rather than a measurement by proxy. While the gluon or quark jet is well defined at the level of the hard scattering, corresponding to an initiating parton of the primary vertex as was discussed above in sect. ?? [9] [7], this definition has limited realizability in experimental searches, and relies on an event labeling that is not supported by Quantum Field Theory approaches [8]. The determination of relative populations of quark and gluon jets in a dijet sample is one of the primary tasks of this thesis and will be discussed at length in chapters 10 and 11.

Kimiske et al.—from here referred to as KMT—set their work in the context of discerning the quark-gluon jet populations in samples rather than applying an event-by-event labeling scheme. KMT’s work begins with the consideration of two samples of data with jets at a fixed p_T range with the condition that one be gluon enriched and one quark enriched. In the specific test case presented in KMT, a Z-jet sample (gluon enriched) and a γ -jet sample (quark-enriched) are used.

1.2 Jet Substructure Measurements

1.3 N-Point Energy Correlator

While the energy correlator has been well defined as an energy flow operator over the celestial block expansion in e^+e^- collisions, such a definition has escaped the case of hadron colliders, given the complications inherent in representations of the initial state of a collision of a composite object.

Part I

Hardware: The Detector and Simulations

Chapter 2

The sPHENIX detector

The sPHENIX detector is part of the Relativistic Heavy Ion Collider (RHIC) complex at Brookhaven National Laboratories (BNL) in Upton, New York. RHIC is one of two major circular particle colliders in the world, the other of which is the Large Hadron Collider at CERN in Geneva, Switzerland. Over the course of the sPHENIX experiment which will run from run 23 until RHIC ceases operations after run 25, RHIC will be colliding gold nuclei and protons with run 23 having been dedicated to commissioning Au+Au running, run 24 dedicated to p+p with three weeks of Au+Au for further tracking commissioning and run 25 dedicated to physics running in Au+Au collisions. RHIC runs with a center of mass energy of $\sqrt{s} = 200\text{GeV}$ for both protons and gold, and additionally is the sole collider that collides polarized protons, which allows for more in depth studies of spin physics. At RHIC, sPHENIX's forerunner, the Pioneering High Energy Nuclear Interaction eXperiment (PHENIX), performed the first measurement of the Quark Gluon Plasma droplets [10] in nuclei-nuclei collisions.

sPHENIX is designed to offer significant upgrades to PHENIX, specifically providing increased coverage with hadronic and electromagnetic calorimeters, increasing effectiveness in studies of jets and heavy flavor at mid-to-high Bjorken x [11][12][13] Broadly, sPHENIX can

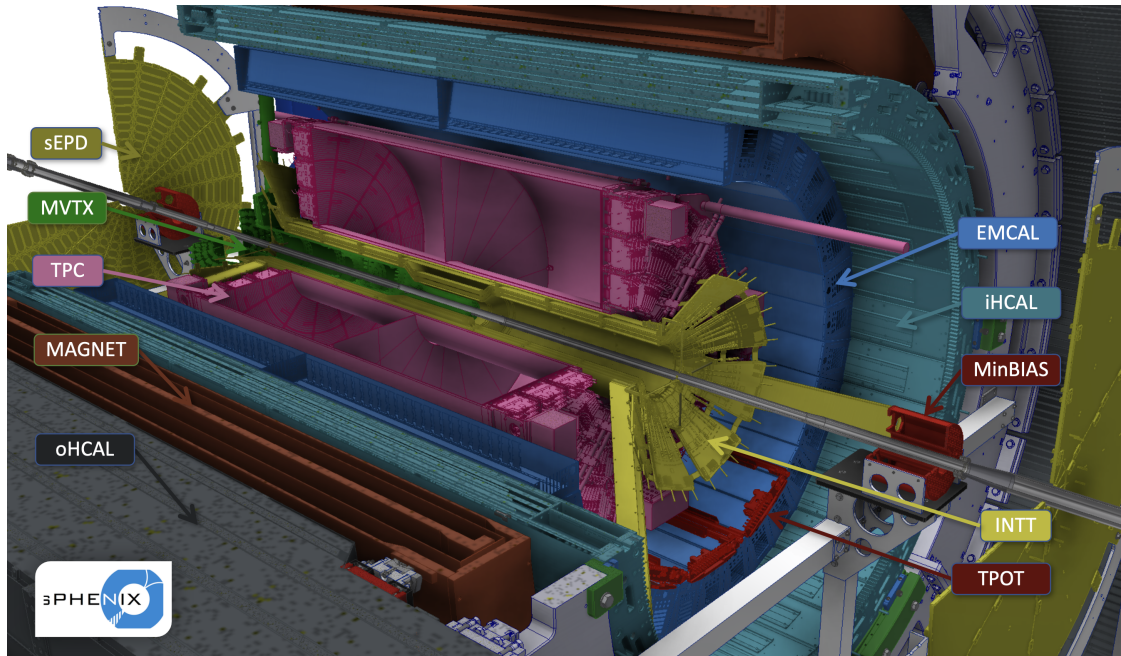


Figure 2.1: The sPHENIX detector. The Zero Degree Calorimeter is not pictured but is located further along the beam pipe.

be broken into three categories of subsystem: Calorimeters, Tracking and Event characterization.

sPHENIX is constructed around a 1.5 Tesla superconducting magnet, cooled by liquid helium, which was previously used on the BABAR experiment at SLAC, with the outer hadronic calorimeter sitting on the outer radius of the magnet, as see in fig 2.1

2.1 Calorimetry

2.1.1 Analog Digital Converters and Data Acquisition

Beyond the purposes and hardware of the detector itself, one can further characterize the subsystems of sPHENIX into streaming readout and non-streaming detectors, broadly distinguishing the tracking detectors from other subsystems. The non-streaming detectors all use a common data acquisition pipeline and data format. These detectors are readout to

analog digital converters boards with 64 channels each, which are then grouped into sets of 3 called an "XMIT" group, which is then read to the data format as a "packet" of 192 channels. Each packet is transmitted from the detector to the computing center on a separate fiber optic cable that is then read into a digitizer which forms groups of (at most) eight packets that are then assigned to individual data acquisition servers. The digitizing process allows for readback into the detector through a global trigger and timing module, where each DAQ server, can issue commands and readback to an event building logic that will then flow back to the detector controlling the same XMIT groups. Each set of (at most) 4 XMIT groups is arranged into a crate with a single crate controller that receives trigger and timing information distributed via a global timing and level 1 trigger module (GL1/GTM) that is then processed through a clockmaster for each digitizer rack, which will hold between 1 and 4 crates, depending of the needs of the subsystem. The data read out to each DAQ server is then stored in the "Purschke Raw Data Format" or prdf file that provides an event structure consisting of a header with some identifying information and separate headers for each packet, but is otherwise comprehensible as an ASCII file. This event structure allows for easy offline combination of events across the calorimeters while keeping individual file size buffers relatively small, as each packet has timing information along with an event number relative to the run that is inherited from the timing module. This DAQ configuration is currently used for the calorimeters, the event plane detector and the minimum bias detector, which allows for faster event building to verify data quality and alignment. The tracking detectors utilize a streaming readout format that creates event pools that are then associated with beam crossing timings provided by the RHIC clock and can be matched onto the GTM timing in offline analysis. This process is slow, and the fully implementation of event combination is still under development, not accounting for event reconstructions which involves track matching across subsystems and will be discussed in section 2.2.5.

ADC Zero Suppression

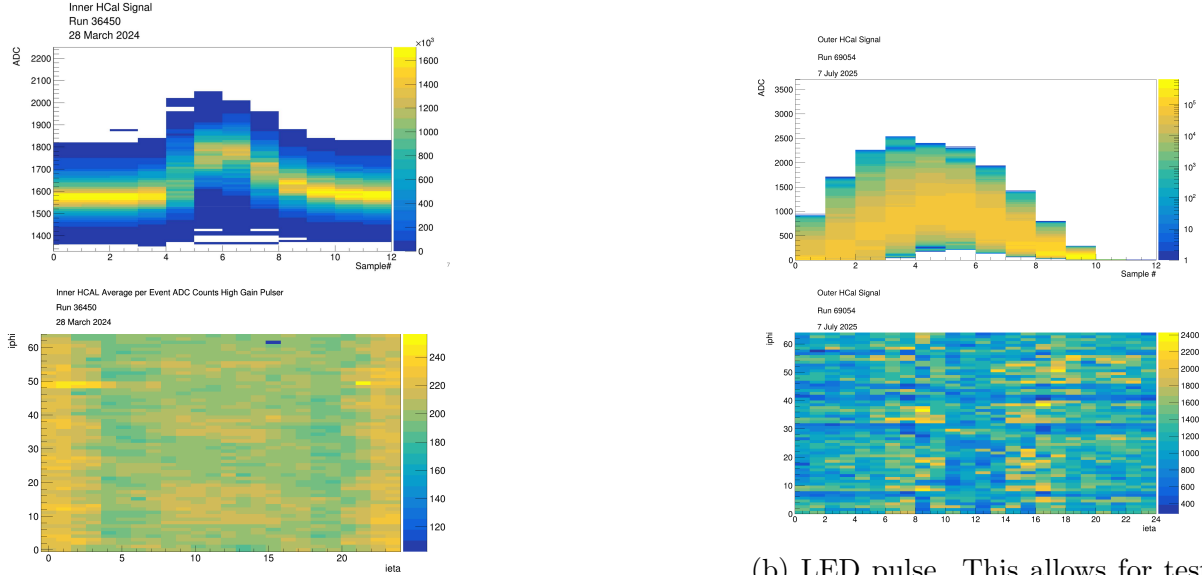
While this system allows for a simplified readout of the calorimetry system, one expects that, for the average event, most of the towers will be dominated by noise, thus one wastes significant computational resources on reading and storing data that will be immediately discarded. This decreases readout rates, runs the risk of filling the storage systems and increases the likelihood of register errors that require restarting runs; in summary, these null data points give no useful physics and hurt our ability to consistently take data. The ADC and readout electronic systems are thus equipped with a zero suppression system that allows for onboard discrimination and data reduction. At the FPGA level, each channel is individually evaluated by taking a post-pre measurement and comparing to a threshold derived from studies of average noise in each channel

$$ADC_{\text{sample 6}} - ADC_{\text{sample 0}} > E_{\text{cut off}} \quad (2.1)$$

Where the evaluation shown in eq (2.1) assumes that the ADC readout has been properly timed in relative to the trigger such that the peak of well formed data should be centered near sample 6 (of 12, 16 or 31), and one can take a cutoff with resolution as wide as a single number and as granular as a tower-by-tower volume. Studies have been performed at sPHENIX to establish good thresholds for this zero-suppression approach and validate that this will not introduce biasing into the detector [14].

2.1.2 Hadronic Calorimeters

sPHENIX's power in jet physics comes in large part from its calorimeter systems, including two separate hadronic calorimeters. The outer and inner hadronic calorimeter (OHCAL and IHCAL respectively) are both divided into 24 bins in η with coverage of $|\eta| \leq 1.1$ and full φ coverage with 64 bins in φ , grouped into 32 sectors across η in each calorimeter. Each



(a) Charge Injection Pulse. This system allows for direct testing of electronics without detector response

(b) LED pulse. This allows for testing of the full readback system and light response

Figure 2.2: Top: Persistence Waveform. Bottom: Energy distribution averaged across 10^5 events

hadronic calorimeter therefore has tower size of $\delta\eta = 0.092$ and $\delta\varphi = 0.098$. The outer hadronic calorimeter is constructed of steel and sits outside of the magnet, with inner radius of $r = 1.82$ m and outer radius of $r = 2.69$ m. The inner hadronic calorimeter is constructed of aluminum and sits just inside the magnet, with inner radius of $r = 1/16$ m and outer radius of $r = 1.37$ m. Each tower corresponds to a readout board that sums readout from four scintillator paddles in the inner HCAL and five scintillator paddles in the outer HCAL embedded into the calorimeter as show in fig. ???. These interface boards form a single readout channel, and can be individually tested via a pulser system that injects charge into the interface board testing readback independent of scintillator response, and LED system that injects a fixed pulse of light into the scintilators to test behavior of the Silicon Photomultipliers (SiPMs). Output of each of these systems is shown in fig 2.2

Calibration

The hadronic calorimeters were initially calibrated through use of cosmic ray muons, matching the spectra to Monte Carlo generated through EcoMug and GEANT4 [**HCal**·**Calib**]. This calibration is updated throughout the run via continued cosmic ray studies in between physics data taking, in addition to ongoing work on in-situ calibrations using tower-slope methods [**tower**·**slope**·**hcal**]. These calibrations have yielded an average conversion factor for the OHCAL of and the IHCAL of

2.1.3 Electromagnetic Calorimeter

Moving in one layer from the hadronic calorimeter, the Electromagnetic Calorimeter (EMCAL) has the same coverage in $\eta - \varphi$ space, but has 16 times as many towers with 96 bins in η and 256 bins in φ . The EMCAL is constructed of blocks of tungsten with embedded scintillating fibers, and has inner radius $r = 0.9$ m and outer radius of $r = 1.16$ m. Similarly to the HCals, the EMCAL is also equipped with a pulser and LED, although the increased number of towers and higher variability in response requires that different pulse widths be used for separate sets of towers to prevent saturation and clipping on the LED pulse.

Calibration

The EMCAL was calibrated to the π^0 mass through the $\pi^0 \rightarrow \gamma\gamma$ decay channel with additional corrections applied via the tower slope. The mean m_{π^0} and width are used as quality assurance plots to monitor radiation damage to the EMCAL on an ongoing basis and provide updated calibrations throughout the runs.

2.1.4 Zero Degree Calorimeter

The Zero Degree Calorimeter (ZDC) is a small transverse energy hadron detector that positioned outside of the experiment hall, in the tunnels along the beam pipe, 18 m from the interaction point on both side. The ZDC is used to detect spectator neutrons to allow for the calculation of multiplicity and event geometry, making it an overlapping detector between the calorimeters and the event characterization [15]. In addition, the ZDC provides a good proxy for total collision rates, useful for both real time monitoring of beam conditions and The sPHENIX ZDC is one of two detectors that was preserved from PHENIX, with minor repairs and upgrades to improve running. In pp collisions, the ZDC provides additional utility as a spin momentum detector that allows for measurement of the spin patterns of the RHIC beam through the Vernier Scan. This functionality is pivotal for the development of sPHENIX's spin physics program in pp and its extensions into the continuing physics goals at the Electron Ion Collider (EIC)[15]. For the purposes of this work, the ZDC is not utilized for either spin physics or trigger capabilities and serves primarily to help establish cross-sections for error determination. While this work does not dive into the subject of spin dependence, such topics are a natural extension and present an interesting avenue for future studies in this vein to be done with the sPHENIX Run 24 data set.

2.2 Tracking

The sPHENIX tracking system consists of the Silicon tracking and vertexing detectors and the gas based trackers. These systems are designed for fast readout, precise momentum resolution and secondary vertex detection. However, this thesis does not make use of the tracking data as track to calorimeter matching is still undergoing development at the time of this analysis, thus the tracking detectors are briefly mentioned here but are not expanded upon in detail

2.2.1 Time Projection Chamber

The Time Projection Chamber (TPC) is a gas based detector designed to track charged particles via the use of a drift chamber. The sPHENIX TPC is filled with a mix of gasses comprising CF_4 , argon, and isobutane—the particular admixture being set in the commissioning phase of run 24 to optimize signal-to-noise ratios.

2.2.2 TPC Outer Tracker

Breaking with the established convention, the next detector is found at larger radial distance from the interaction point than the TPC. The TPC Outer Tracker (TPOT) was designed as a minimal add-on to improve tracking efficiency and resolution by providing additional track points.

2.2.3 Intermediate Silicon Tracking Detector

The INTermediate Silicon Tracking detector (INTT) consists of two layers of semiconducting silicon.

2.2.4 Micromegas Vertex Detector

The Micromegas Vertex detector (MVTX) is a silicon based vertexing detector based on a similar system at the ALICE experiment of the Large Hadron Collider (LHC). This detector sits closest into the beam pipe and is instrumental in the open heavy flavour program of sPHENIX [11] The MVTX is comprised of three layers of 8, 12 and 21 staves in concentric rings. Each of these is then further divided into

2.2.5 Event Reconstruction and Track Matching

2.3 Event Characterization

The event characterization detectors are of primary use in AuAu collisions, both sitting at high η and designed to allow for analysis of events through spectator particle output, which, while some persist in p+p collisions, these effects are much more significant in heavy ion collisions. Therefore, the impact of these detectors on this analysis are minimal.

These two detectors fill out the compliment of ADC systems and as such are run in triggered mode in the same manner as the calorimeters, and build events in the same manner as the other ADC systems from the perspective of the data acquisition system and offline event combining.

2.3.1 sPhenix Event Plane Detector

2.3.2 Minimum Bias Detector

The sPHENIX Minimum Bias Detector (MBD) is the other detector preserved from PHENIX, when it was called the Beam-Beam Counter (BBC). This detector is situated at high eta in both north and south ends of the detector, and provides a z-vertex available at the online level. The MBD has two major outputs, measuring charge and time of particles passing through the detector

2.4 Triggers

The sPHENIX detector has a number of triggers at its disposal to allow for online event type selections for the triggered readout system: the EMCal, the HCals, the ZDC, the MBD, the sEPD and—for run 2024—the TPC. Starting in run 2024, available triggers are

ZDC and Minimum Bias in single and coincidence modes, a restricted vertex Minimum Bias trigger, four jet triggers at differing minimum energy thresholds (6-12 GeV with variation through the run), four photon triggers at differing minimum energy thresholds (2-10 GeV with variation through out the run), random triggers, clock triggers and jet with Minimum Bias triggers. In order to decrease busy signals, the overall trigger rate is scaled down to 8 kHz, taking 100 Hz of clock triggers as a control, full rate rare event (jet and photon) and minimum bias filling the remaining.

2.4.1 Jet Triggers

2.4.2 Photon Triggers

2.4.3 Minimum Bias Trigger

Chapter 3

Monte Carlo Simulations

For proton-proton collisions, sPHENIX has 10^8 fully reconstructed events from Pythia 8 and Herwig 7 for $2 \rightarrow 2$ and minimum bias (soft QCD) events that are then reconstructed using GEANT. While there is a jet sample that has been performed with the Detroit tune [16] and there is an existing, equivalent tune for Herwig [17], in this work, I focus on the default tuned sample as there is a wider dataset from which to draw. data from the 2 and three point energy correlator, and its important for training of our discriminator

435 Chapter 4

436 Determining Backgrounds and Errors

Part II

Technicalities: The finer points of theory and computing

Chapter 5

The Energy Correlator and the primary vertex

Considering the limitations of Calorimetry only measurement with RHIC one is So the energy correlator allows a look back, I guess that's the relevant bit? Point being, if the jet points back to the primary vertex, the share of the energy between the jets and the composition of those jets is directly determined by the $2 \rightarrow 2$ scattering and is thus sensitive to the parton distribution functions of the beam particles. Starting from the definition of the energy correlator given in [18]:

$$\mathcal{E}_N = \int \frac{d\Omega}{2} \int d\vec{n}_{12} \frac{\langle \varepsilon(\vec{n}_1) \dots \varepsilon(\vec{n}_2) \rangle}{Q^2} \delta(\vec{n}_1 \cdot \vec{n}_2 - 2|\vec{n}_1||\vec{n}_2| \cos \theta) \quad (5.1)$$

Where $\varepsilon(\vec{n}_i)$ is the energy flow operator defined as

$$\varepsilon(\vec{n}_i) = \lim_{r \rightarrow \infty} r^2 \int_{-\infty}^{\infty} dt \vec{n}^i T_i^0(t, r\vec{n}) \quad (5.2)$$

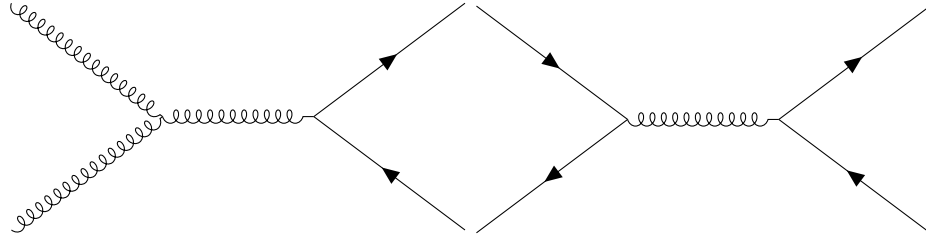


Figure 5.1: $p + p \rightarrow q\bar{q} + X$ events, these dijet-types are possible to be produced from gluon or quark interchange, probing the square of the gluon or quark distributions

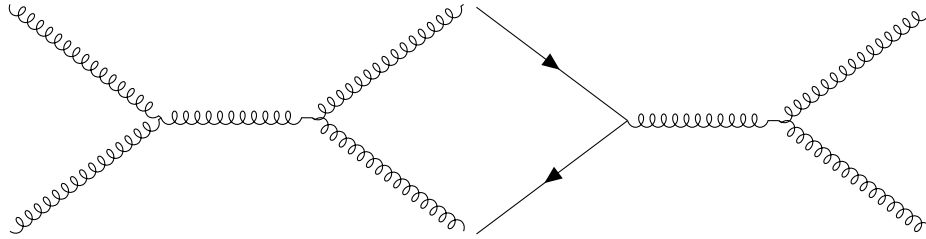


Figure 5.2: $p + p \rightarrow gg + X$ events, these dijet-types are possible to be produced from gluon or quark interchange, probing the square of the gluon or quark distributions

5.1 The primary vertex and interaction

The goal of this work is to extract gluon distributions from the jet, thus one must consider those initial vertices that (to first order) produce the gluon-quark, gluon-gluon, and quark-quark dijet events. To this end, one considers the following tree level diagrams. While the processes are not immediately apparent in the final state of the jet, one notes that the two and three point energy correlators give a useful distinction as the jet function can be divided out, leaving a vertex dependent term at $R_L \approx 2R_{\text{jet}}$

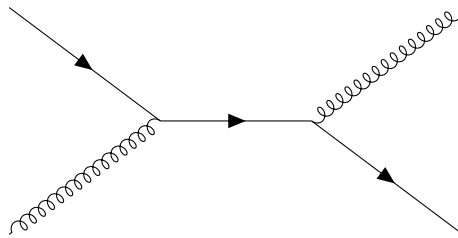


Figure 5.3: $p + p \rightarrow gq (\bar{q}) + X$ events, these dijet-types are possible at the tree level only via quark (antiquark) and gluon interactions, hence probing both gluon and quark distributions at once

Chapter 6

Jets in Vacuum and the PDF

6.1 Jet Identification Algorithms

As discussed in chapter 1, there are a variety of jet finding algorithms that prioritize different theoretical aspects of the underlying physics while being experimentally realizable [3] [1].

In general, a jet identification algorithm needs to be IRC safe. That is, the jet object needs to display invariance in the Infrared (IR) and Collinear regimes, managing real-virtual cancellation and keeping results meaningful for emission and splitting respectively.

465 Chapter 7

466 So exactly how intelligent is AI

Part III

Experimental Output: The Main Event

470 Chapter 8

471 So is sPHENIX actually working?

472 Chapter 9

473 Measuring the Energy Correlator

474 9.1 Discretization and Efficiency

9.2 Efficiency corrections

As was discussed in section ??, the selection of a jet cone limits the accuracy of the energy correlator calculation in a way that depends on R_L . Beyond the problems noted in that model, one also has the issue of normalization and efficiency. Any set of pairs (or triplets or N-point combinations) measured in a calorimeter needs to be corrected for the overabundance of pairs sitting at the discretized lattice in R_L . To proceed with the efficiency correction on the two point correlator, one must begin with considering the continuous case. Any jet cone limits the number of identified pairs which has the potential to bias the results, thus an efficiency of identification of contained pairs must be introduced. Begin by drawing a jet cone of radius R_j in the $\eta - \varphi$ plane of some idealized zero-width tower calorimeter, and select some point in the jet offset from the center by some distance and angle λ , then draw another circle of radius R_L around this point limited to the jet cone—representing all pairs to the chosen point that lie within the jet—as show in fig ??. Then the efficiency ratio defined as geometric pairs in jet over geometric pairs in the phase space gives

$$\begin{aligned} \tilde{\varepsilon}(R_L, R_J) &= \frac{\int_0^{2\pi} d\lambda \int_{\max\{0, R_L - R_j\}}^{R_j} d\tau C_{inc}}{C_{Total}} \\ &= \frac{\int_0^{2\pi} d\lambda \int_{\max\{0, R_L - R_j\}}^{R_j} d\tau 2 \int_0^{\theta_f} d\theta \pi \theta R_L}{\int_0^{2\pi} d\lambda \int_0^{R_j} d\tau 2\pi R_L} \end{aligned} \quad (9.1)$$

Where θ is the polar angle measured around the circle of pairs, C_{inc} is the circumference of the circles of pairs, C_{Total} is just the total pairs in the phase space and is used as a normalization factor to allow for comparisons with the discretized case. The bounds for the θ integral reflect that two points are symmetric around the axis defined by the centers of the two circles, and $\cos \theta_f = \frac{R_L^2 + \tau^2 - R_j^2}{2\tau R_j^2}$, and the bounds on τ are given to keep at least one pair within the jet in the numerator. Switching the order of integration in eq (9.1) and bounds allows for a version of this integral to be done, and then the resulting elliptical integral is

496 evaluated around $\arccos\left(\frac{R_L}{2R_j}\right)$ for an approximate answer

$$\begin{aligned}\tilde{\varepsilon}(R_L, R_j) &= \frac{R_L}{2\pi R_j^2} \sqrt{4R_j^2 - R_L^2} + \frac{E\left(\arccos\left(\frac{R_L}{2R_j}\right) \middle| \frac{R_L^2}{R_j^2}\right)}{2\pi} \\ &\approx \sqrt{R_L} 2\pi R_j^2 \left[\frac{4R_j^2 - R_L^2}{+} R_j \arccos\left(\frac{R_L}{2R_j}\right) \right. \\ &\quad \left. - \frac{R_L}{3} \left(\arccos\left(\frac{R_L}{2R_j}\right)\right)^3 + \mathcal{O}\left(\left[\arccos\left(\frac{R_L}{2R_j}\right)\right]^5\right) \right]\end{aligned}\quad (9.2)$$

497 Switching to the discretized case, one introduces the new parameters δ, γ being the
498 size of a calorimeter tower in φ, η respectively. First take $\delta \approx \gamma$ such that the problem is one
499 of square towers, which, for the purposes of analysis of sPHENIX data is appropriate, given
500 that the towers of each calorimeter are almost square—see sections ?? and ??. Then, we start
501 with a definition of the circumference to use in this case. One states that all towers can be
502 expressed as a set of points of the form $(p\delta, q\delta)$ s.t. $p, q \in \mathbb{Z}$, and the equivalent measure to
503 the circumference of the continuous around a central point $P_c = (p_c, q_c)\delta$ can be stated as

$$|\mathcal{C}[R_L, \delta, (p_c, q_c)]| = \left| \left\{ P \mid (p - p_c)^2 + (q - q_c)^2 = \frac{R_L^2}{\delta^2} \right\} \right| \quad (9.3)$$

504 A natural extension from this measure tells us that $\frac{R_L^2}{\delta^2} = \mathbb{Z}$ which gives a natural binning
505 for the energy-energy correlator, if plotted as a function of R_L^2 . From here, the equivalent
506 expression to eq. 9.1 can be stated as

$$\tilde{\varepsilon}(R_L, R_j, \delta) = \frac{\sum_{(m,n) \in \text{Disk}(R_j)} |\mathcal{C}[R_L, \delta, (m, n)] \cap \text{Disk}(R_j)|}{|\text{Disk}(R_j)| |\mathcal{C}[R_L, \delta, (0, 0)]|} \quad (9.4)$$

507 where $\text{Disk}(R_j)$ is the collection of points that lie within a circle of radius R_j and $\mathcal{C}$ is the
508 collection of points at exactly R_L from the center point $(m, n)\delta$. The cardinality of these
509 sets can be found by noting that $|\mathcal{C}|$ is simply the sum-of-squares function $r_2\left(\frac{R_L^2}{\delta^2}\right)$ and

510 $|\text{Disk}(R_j)|$ is Gauss's Circle Problem [19].

511 In order to solve for the numerator, one takes a second disk of radius R'_j such that,

$$\mathcal{C}[R_L, \delta(m, n) \cap \text{Disk}(R'_j)] = \mathcal{C} \quad \forall (m, n) \in \text{Disk}(R_j) \quad (9.5)$$

512 Which, allows for the folding of the summation into a problem of set theory, noting that

$$\begin{aligned} \text{Disk}(R'_j) &= \text{Disk}(R_j) \cup \sum_{(m,n)} \mathcal{C} \\ \Rightarrow |\text{Disk}(R_j) \cap \sum_{(m,n)} \mathcal{C}| &= |\text{Disk}(R_j)| + \left| \sum_{(m,n)} \mathcal{C} \right| - |\text{Disk}(R'_j)| \\ &= |\text{Disk}(R_j)| (1 + |\mathcal{C}[R_L, \delta, (0, 0)]|) - |\text{Disk}(R'_j)| \end{aligned} \quad (9.6)$$

513 Setting $R'_j = R_j + R_L$ as the largest value for R'_j which corresponds to a collection of pairs
514 centered on one of the edges of the faces, the equivalent to eq 9.2 is simply

$$\tilde{\varepsilon}(R_L, R_j, \delta) = 1 - \frac{1}{r_2\left(\frac{R_L^2}{\delta^2}\right)} \left[\frac{\left(\frac{(R_j+R_L)^2}{\delta^2}\right)^i}{\sum_{i=\frac{R_j^2}{\delta^2}}^{\frac{R_L^2}{\delta^2}}} r_2(i) \bigg/ \sum_{i=0}^{\frac{R_j^2}{\delta^2}} r_2(i) \right] \quad (9.7)$$

515 To approximate the solution one can the identity from Hardy and Ramanujan [20], however,
516 for our purposes, it suffices to use numerical solutions for the special cases that are relevant
517 to the detector as is shown in fig ?? Thus, the full efficiency correction on ΔR_L to correct
518 for both edge effects of the jet cone and the effects of a finite spatial resolution on the

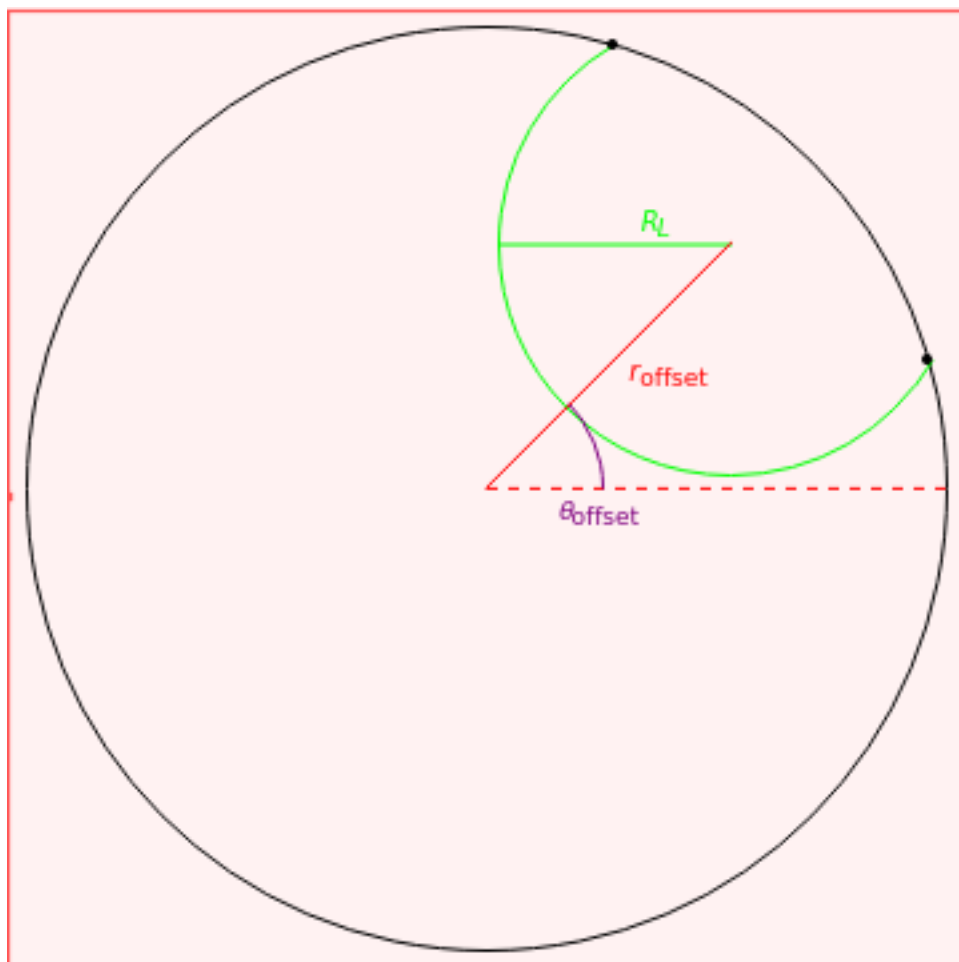


Figure 9.1: The continuous and discretized calorimeters for phase space and discretization efficiency correction to smooth distribution and reduce bin width selection dependence

519 calorimeters is

$$\varepsilon(R_L, R_j, \delta) = \frac{R_L r_2\left(\frac{R_L^2}{\delta^2}\right)}{2\pi R_j^2} \left[\frac{\sqrt{4R_j^2 - R_L^2} + R_j E\left(\arccos\left(\frac{R_L}{2R_j}\right) \middle| \frac{R_L^2}{R_j^2}\right)}{r_2\left(\frac{R_L^2}{\delta^2}\right) - \sum_{i=\frac{R_j^2}{\delta^2}}^{\frac{(R_j+R_L)^2}{\delta^2}} r_2(i) \bigg/ \sum_{i=0}^{\frac{R_j^2}{\delta^2}} r_2(i)} \right] \quad (9.8)$$

520 **9.3 Uncertainties**

521 **9.3.1 Statistical Uncertainty**

522 **9.3.2 Systematic Uncertainty sources**

523 **9.3.3 Binning Uncertainty**

524 **9.3.4 Systematic Uncertainties**

In the discussion of the energy correlator in chapter 5, we had been working in an idealized regime where the measurement is able to be resolved to arbitrary size, however, this is limited by the physical reality of the detector systems.

In simulations and data, one finds that the practical reality of making measurements of the Energy correlator is severely hamstrung by the use of tower size objects rather than the truth particles that make up our idealized toy jets discussed in section 6.1. To study the efficiency loss, one takes the identified truth jet objects from Monte Carlo Simulations (Pythia8 and Herwig for pp; sHIJING, EPOS4, AMPT for AuAu) and runs them through Geant simulations to model detector effects. One then takes the initial truth jet, draws a circle that encloses the truth particles, and projects this out onto the calibrated tower response in each calorimeter, as well as runs an anti- k_T jet identification algorithm to simulate jet reconstruction. These three object types (truth jet, projective cone and reconstructed jet) are collected into sets and the measurement of the two- and three-point energy correlators is performed using the definition in eq 9.9

$$\langle E2C(R_L) \rangle = \frac{1}{N_{\text{events}}} \sum_{\text{events}} \frac{1}{N_{\text{jet}}} \sum_{N_{\text{jets}}} \sum_{i,j \in \text{Jet}} \sum_{\text{Consitutents}} \frac{E_i E_j}{E_{\text{jets}}^2} \delta(R_{ij} - R_L) \quad (9.9)$$

$$\langle E3C(R_L) \rangle = \frac{1}{N_{\text{events}}} \sum_{\text{events}} \frac{1}{N_{\text{jet}}} \sum_{N_{\text{jets}}} \sum_{i,j,k \in \text{Jet}} \sum_{\text{Consitutents}} \frac{E_i E_j E_k}{E_{\text{jets}}^3} \delta(\max R_{i,j,k} - R_L)$$

9.4 Whole Calorimeter Measurements

While the Energy correlator is generally studied within a jet, one can make use of the whole calorimeter to compare the energy correlator across all angular separations to characterize the underlying event. This induces a new computational issue of scale, where the number of calculations goes from $_{50}C_2 = 1225$ in the HCAL and $_{804}C_2 = 322,806$ in the EMCAL to $_{1536}C_2 \approx 2.4 \times 10^6$ and $_{24576}C_2 \approx 6 \times 10^8$ in the EMCAL. However, since pp events

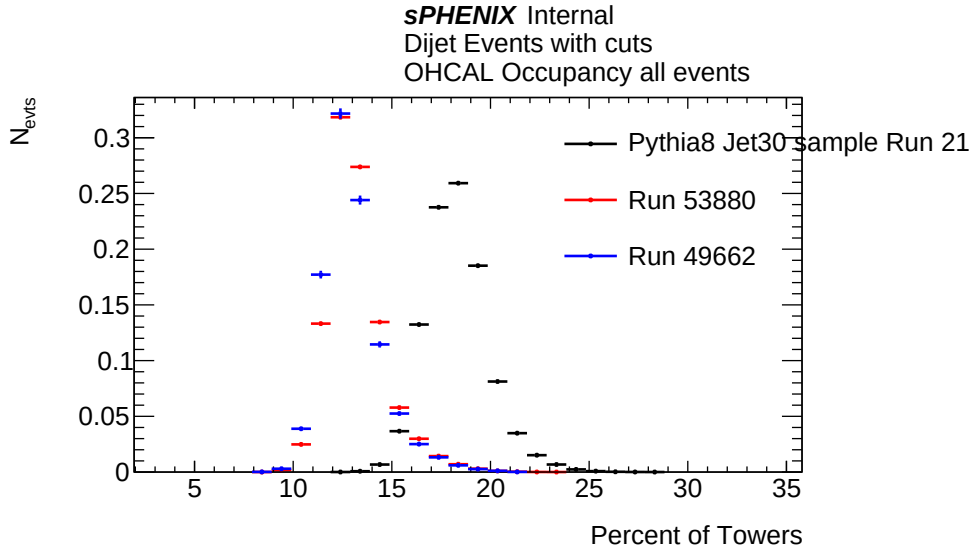


Figure 9.2: Multiplicity in each Calorimeter (pending–have ohcal now) as percentage of towers with $E \geq 10$ MeV to filter out electronics fluctuations

have relatively low multiplicity in the calorimeters as is seen in fig 9.2, one reduces the computational load significantly by preprocessing the towers to remove all towers with energy below some threshold, functionally mirroring the zero-suppression procedure discussed in section 2.1.1.

One also notes that the whole event correlator on calorimeter towers is significantly sensitive to the electronic noise of the towers, given that any positive electronic noise would mimic a valid energy deposition. While the zero suppression reduces the noise by a significant fraction at the hardware level, there are fluctuations that pass this cut. Each event has an entry for every tower, and due to this electronic noise not having the same waveform shape in the ADC (see figs ?? and ?? for a comparison of a pedestal waveform and that of a data event–there are a fraction of towers that have an apparent negative energy, which when summing the total energy of the calorimeters, approximately cancels out positive fluctuations on average. However in our case, it is not so simple. Consider the following ansatz of a symmetric distribution $S = \{-2, -2, -1, -1, -1, -1, 1, 1, 1, 1, 2, 2\}$, then one considers the sum of the product of pairs, being the numerator of the two point energy correlator would

560 be the only way to have the noise contribution go to zero

$$\begin{aligned}
 PS &= (-2)^2 + (-2 \cdot -1) \times 4 + (-2 \cdot 1) \times 4 + (-2 \cdot 2) \times 2 \\
 &\quad (-2 \cdot -1) \times 4 + (-2 \cdot 1) \times 4 + (-2 \cdot 2) \times 2 \\
 &\quad + (-1)^2 \times (3 + 2 + 1) + (-1 \cdot 1) \times 4 \times 4 + (-1 \cdot 2) \times 4 \times 2 \\
 &\quad + (1)^2 \times (3 + 2 + 1) + (1 \cdot 2) \times 4 \times 2 \\
 &\quad + 2 \cdot 2 \\
 &= -4 + -8 + -26 + 22 + 4 \\
 &= -12
 \end{aligned} \tag{9.10}$$

561 Clearly a symmetric distribution alone does not guarantee a full cancellation. For a full
 562 proof of this for a Gaussian distribution see appendix A.

Chapter 10

The Power of the ENC: α_s at the few GeV scale

As had been shown recently at CMS [21], taking the ratio of the two and (projective) three point energy correlator results in a distribution that scales directly with $\alpha_s(Q^2)R_L$ where Q^2 is the scale of the process under consideration. To make this calculation in the measurement presented in this thesis, we need to fix the hard scale relevant to the event. Here it is instructive to consider how we can adapt the methods utilized for a jet based energy correlator measurement to those for an event based model. In the CMS analysis is [21], they use a definition of the perturbative scale and hard scale (where $x_L \equiv R_L$ in this thesis) as

$$\begin{aligned} PS &= \frac{p_T^{jet} R}{2} \\ Q^2 &= (ap_T^{jet} x_L)^2 \end{aligned} \tag{10.1}$$

Chapter 11

Event Classification

The more standard method of measuring α_S requires that one be able to accurately determine the relative quark and gluon jet populations in the event samples [22]. While the discussion in chapter 10 allow for us to circumvent this issue in the immediate measurement of the strong force coupling, it is still a useful quantity to measure for further studies on jet substructure measurements. So while it is inappropriate to tag individual jets or events as a "quark" or "gluon" type event through the Energy Correlator, given that any individual jet may have fluctuations that overlap into

583 Chapter 12

584 Comparison is the Theft of Joy: What
585 does the LHC say, and how about
586 STAR?

587 Chapter 13

588 Entanglement and other Lofty Goals

589 In this thesis, I have presented a limited scope measurement of a whole event energy correlator
590 in p+p event which still empowers us to use it to measure α_S and to estimate our gluon and
591 quark jet populations.

592

Part IV

593

Wrapping it all up

Chapter 14

Observable prospects for p+p in run

24, AuAu with run 24 and 25 and the

EIC

598 Chapter 15

599 Remaining Questions

600 Chapter 16

601 Implementation and application for
602 the remaining sPHENIX data

603

Part V

604

Odds and Ends-Matter

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Appendices

Appendix A

Proof that a Symmetric Energy Distribution fails to cancel in the Product-Sum

In the ansatz reflected in equation (9.10), a symmetric set with a distribution roughly approximating a Gaussian was utilized, however, we can go beyond that to properly show this result is the case in a set drawn from a generalized Gaussian distribution

Proof

We begin with a Gaussian distribution $P(x)$ with amplitude A and sample points from it such that we construct a "Gaussian sample set" with the following properties

Definition A.0.1 (Gaussian Sample Set). Take a set S that is constructed of points randomly sampled such that

$$S \approx P(x) = \frac{A}{\sqrt{2\pi}\sigma} e^{\frac{-x^2}{\sigma}} \quad (\text{A.1})$$

Which gives the following properties

1. $S = \{x | \bar{x} = \delta_{\bar{x}}\}$
2. $\frac{1}{N} \sum_{x \in S} x^2 = (\sigma + \delta_{\sigma})^2$
3. $|\delta_{\bar{x}}|, |\delta_{\sigma}| \ll 1$
4. $\lim_{N \rightarrow \infty} |\delta_{\bar{x}}|, |\delta_{\sigma}| = 0$

Then, the Product-Sum is calculated by following the standard algorithm

Algorithm: PRODUCT-SUM

$$\begin{aligned}
& \forall x \in S \text{ s.t. } \text{len}(S) > 1 \\
& S = S/x \\
& PS = PS + (x * y) \quad \forall y \in S
\end{aligned} \tag{A.2}$$

Product-sum algorithm for the sample set S. Initially the PS variable is set at 0

Using the properties of S noted in definition A.0.1, one notes that for any sampled value x, there should be approximately $n = \left\lfloor \frac{A}{\sqrt{2\pi(\sigma+\delta\sigma)}} e^{\frac{(x-\delta_{\bar{x}})^2}{\sigma+\delta\sigma}} \right\rfloor$ of the same value. In fact, by including these small corrections and an additional small correction $|\delta_{n,x}| \ll 1$ we fix this number to the Gaussian curve within some fluctuations. With this in mind, one breaks the contributions to the Product-sum into two categories, the $y = x$ and $y \neq x$ case. For the first case, the contribution is simply

$$PS_{y=x} = x^2 (n(x) - 1)! \tag{A.3}$$

and in the second case

$$PS_{y \neq x} = xyn(x)n(y) \tag{A.4}$$

Thus the product-sum becomes

$$PS = \sum_{x \in S} \left[x^2 (n(x) - 1)! + \sum_{y > x, y \in S} xyn(x)n(y) \right] \tag{A.5}$$

With a sufficiently high number of samples, we can shift to the continuum limit and write

$$\widetilde{PS} = \int_{-x_0}^{x_0} dx \left[x^2 \Gamma((n(x))) + \int_x^{x_0} dy \frac{xy}{2\pi(\sigma + \delta\sigma)} e^{\frac{-(x-\delta_{\bar{x}})^2 - (y-\delta_{\bar{x}})^2}{2(\sigma+\delta\sigma)}} \right] \tag{A.6}$$

where x_0 is a limit set by finding the point at which there is only 1 measurable point

$$n(x_0) = 1 \Rightarrow x_0 = \delta_{\bar{x}} + \sqrt{2(\sigma + \delta\sigma) \ln A - (\sigma + \delta\sigma) \ln [2\pi(\sigma + \delta\sigma)]} \tag{A.7}$$

685 Taking the part of this integral where of $x \neq y$, one finds

$$\begin{aligned}
\widetilde{PS}_{x \neq y} &= \int_{-x_0}^{x_0} dx \frac{x}{2\pi(\sigma + \delta\sigma)} e^{\frac{-(x-\delta_{\bar{x}})^2}{2(\sigma+\delta\sigma)}} \int_{x_0-\delta_{\bar{y}}}^{x_0} dw (w + \delta_{\bar{y}}) e^{\frac{-w^2}{2(\sigma+\delta\sigma)}} \\
&= \int_{-\infty}^{\infty} dx \frac{x}{2\pi(\sigma + \delta\sigma)} e^{\frac{-(x-\delta_{\bar{x}})^2}{2(\sigma+\delta\sigma)}} \left[(\sigma + \delta\sigma) e^{\frac{-(x-\delta_{\bar{y}})^2}{2(\sigma+\delta\sigma)}} + \delta_{\bar{y}} \int_{x-\delta_{\bar{y}}}^{\infty} dw e^{\frac{-w^2}{2(\sigma+\delta\sigma)}} \right] \\
&= \int_{-\infty}^{\infty} dx \frac{x}{2\pi} e^{\frac{-(x-\frac{\delta_{\bar{x}}+\delta_{\bar{y}}}{2})^2}{\sigma+\delta\sigma} + \frac{(\delta_{\bar{x}}-\delta_{\bar{y}})^2}{4(\sigma+\delta\sigma)}} + \int_{-\infty}^{\infty} dx \frac{x}{2\pi(\sigma + \delta\sigma)} e^{\frac{-(x-\delta_{\bar{x}})^2}{2(\sigma+\delta\sigma)}} \delta_{\bar{y}} \int_{x-\delta_{\bar{y}}}^{\infty} dw e^{\frac{-w^2}{2(\sigma+\delta\sigma)}} \\
&= \frac{1}{2\pi} \frac{\delta_{\bar{x}} + \delta_{\bar{y}}}{2} \sqrt{\pi(\sigma + \delta\sigma)} e^{\frac{(\delta_{\bar{x}}-\delta_{\bar{y}})^2}{4(\sigma+\delta\sigma)}} + \int_{-\infty}^{\infty} dx \frac{x}{2\pi(\sigma + \delta\sigma)} e^{\frac{-(x-\delta_{\bar{x}})^2}{2(\sigma+\delta\sigma)}} \delta_{\bar{y}} \int_{x-\delta_{\bar{y}}}^{\infty} dw e^{\frac{-w^2}{2(\sigma+\delta\sigma)}} \\
&= 0 + \int_{-\infty}^{\infty} dx \frac{x}{2\pi(\sigma + \delta\sigma)} e^{\frac{-(x-\delta_{\bar{x}})^2}{2(\sigma+\delta\sigma)}} \times 0 \\
&= 0
\end{aligned} \tag{A.8}$$

686 Where the last step notes that $\lim_{N \rightarrow \infty} \delta_{\bar{x}}, \delta_{\bar{y}} = 0$ by property 4 of the definition A.0.1, but
687 is not necessarily 0 for a discrete sample set

688 Thus the product sum goes simply to

$$\widetilde{PS} = \int_{-x_0}^{x_0} dx \quad x^2 \Gamma \left(\frac{A}{\sqrt{2\pi(\sigma + \delta\sigma)}} e^{\frac{-(x-\delta_{\bar{x}})^2}{2(\sigma+\delta\sigma)}} \right) \tag{A.9}$$

689 Which is always positive, given that the integrand is always positive as is shown in fig. A.1,
690 thus the product sum is non-zero even in the continuum limit, thus for a Gaussian distribution
691 centered around 0, the sum of two point products is not guaranteed to be positive and thus
692 the symmetry of the distribution can not be exploited to cancel out the noise profile $\mathcal{QED}$

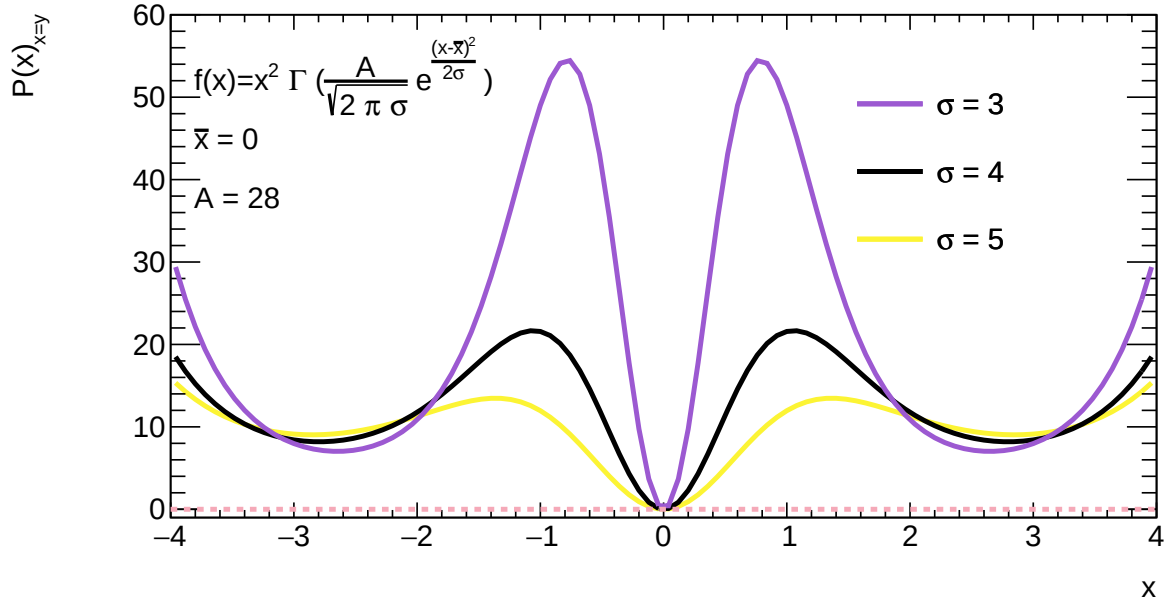


Figure A.1: Select plots of specific cases of the integrand in equation (A.9). The gamma function is always positive for positive real numbers, thus the integrand is always positive and will net a positive result when evaluated